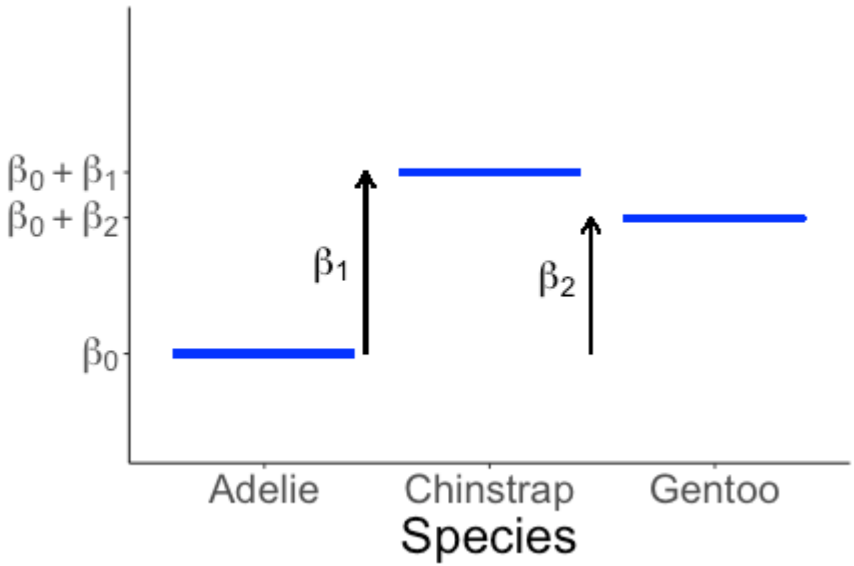
Regression and hypothesis testing with a categorical explanatory variable

Categorical predictor

- How do we model the relationship between species and bill length?
- Can we test for a relationship between bill length and species?

Categorical predictor



$$\text{bill length} = \begin{cases} \beta_0 + \varepsilon & \text{species} = \text{Adelie} \\ \beta_0 + \beta_1 + \varepsilon & \text{species} = \text{Chinstrap} \\ \beta_0 + \beta_2 + \varepsilon & \text{species} = \text{Gentoo} \end{cases}$$

> How can I write this more concisely?

Indicator variables

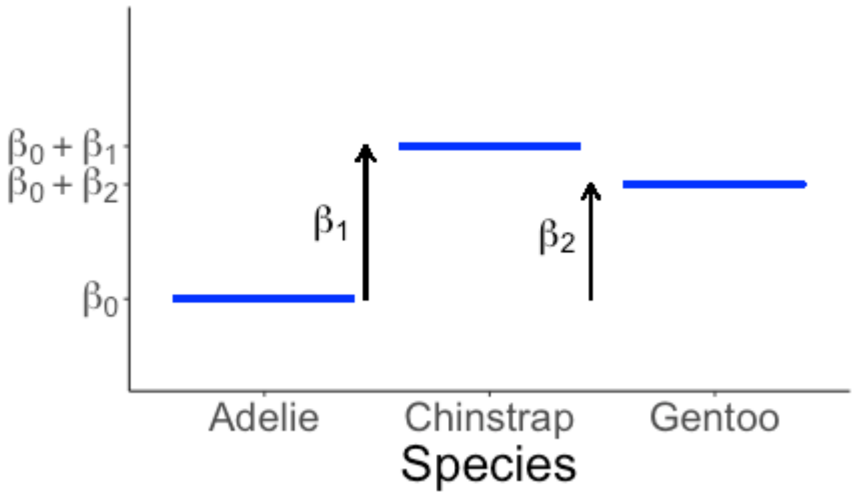
$$\text{IsChinstrap} = \begin{cases} 0 & \text{species} \neq \text{Chinstrap} \\ 1 & \text{species} = \text{Chinstrap} \end{cases}$$

$$\text{IsGentoo} = \begin{cases} 0 & \text{species} \neq \text{Gentoo} \\ 1 & \text{species} = \text{Gentoo} \end{cases}$$

Then:

Species	IsChinstrap	IsGentoo
Adelie	0	0
Chinstrap	1	0
Gentoo	0	1

Indicator variables



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Or, more concisely:

$$\text{bill length} = \beta_0 + \beta_1 \text{IsChinstrap} + \beta_2 \text{IsGentoo} + \varepsilon$$

Model assumptions

$$\text{bill length} = \beta_0 + \beta_1 \text{IsChinstrap} + \beta_2 \text{IsGentoo} + \varepsilon$$

> What model assumptions do you think we make?

- constant variance (variance of ε is the same for each species)
- randomness (data came from random process, random sample, etc.)
- normality (distribution of ε is normal)
- independence (observations are independent)

Model assumptions

$$\text{bill length} = \beta_0 + \beta_1 \text{IsChinstrap} + \beta_2 \text{IsGentoo} + \varepsilon$$

where $\varepsilon \sim N(0, \sigma_\varepsilon^2)$ and the ε are independent of each other.

- No notion of shape assumption with single categorical predictor
- Still assume constant variance, normality, independence, and randomness

think about data generation

residual plot

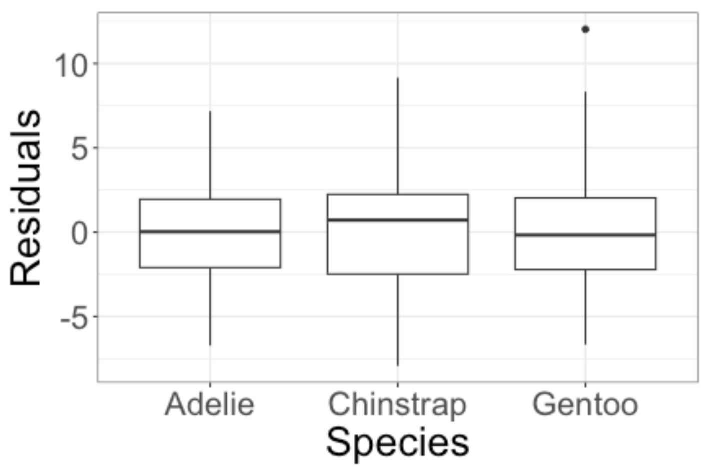
QQ plot

think about data generation

> How should we assess each of these assumptions?

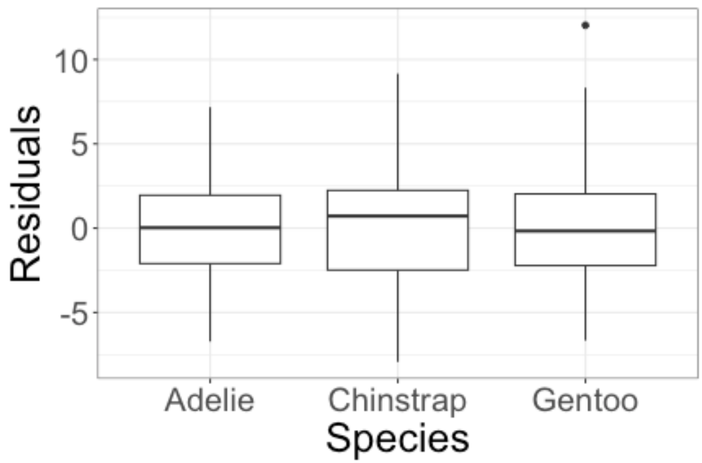
Assessing constant variance

For a single categorical predictor, we can create a boxplot of residuals for each group, and calculate the standard deviation of residuals in each group.



species	residual_std_dev
Adelie	2.66
Chinstrap	3.34
Gentoo	3.11

Assessing constant variance



species	residual_std_dev
Adelie	2.66
Chinstrap	3.34
Gentoo	3.11

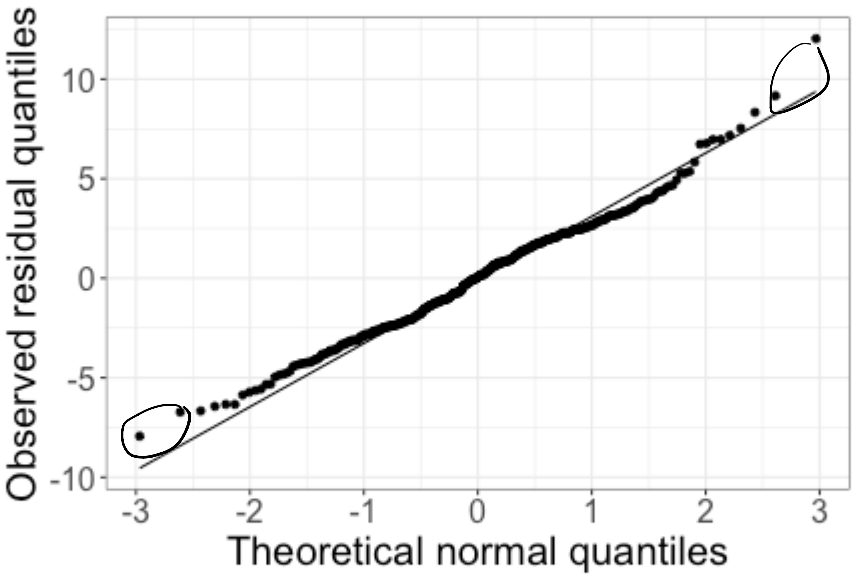
Looking for similar variability between groups. Rule of thumb: not too concerned about constant variance if

$$\frac{\text{largest standard deviation}}{\text{smallest standard deviation}} < 2$$

Here: $\frac{3.34}{2.66} = 1.26 < 2$

Assessing normality

Can use a QQ plot again:



Slight deviation in tails, but overall normality looks good

Statistical inference

> Does it look like there is a relationship between species and bill length?

Hypothesis testing

$$\text{bill length} = \beta_0 + \beta_1 \text{IsChinstrap} + \beta_2 \text{IsGentoo} + \varepsilon$$

We want to know whether there is a relationship between species and bill length.

> Can I translate this into null and alternative hypotheses about one or more model parameters?

H_0: $\beta_1 = \beta_2 = 0$

H_A: at least one of $\beta_1, \beta_2 \neq 0$

Hypothesis testing

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Full and reduced models

$$\text{bill length} = \beta_0 + \beta_1 \text{IsChinstrap} + \beta_2 \text{IsGentoo} + \varepsilon$$

$H_0 : \beta_1 = \beta_2 = 0$

H_A : at least one of $\beta_1, \beta_2 \neq 0$

Reduced model (H_0): model under H_0 (i.e. $\beta_1 = \beta_2 = 0$) (no relation w/ species)

$$\text{bill length} = \beta_0 + \varepsilon$$

Full model (H_A):

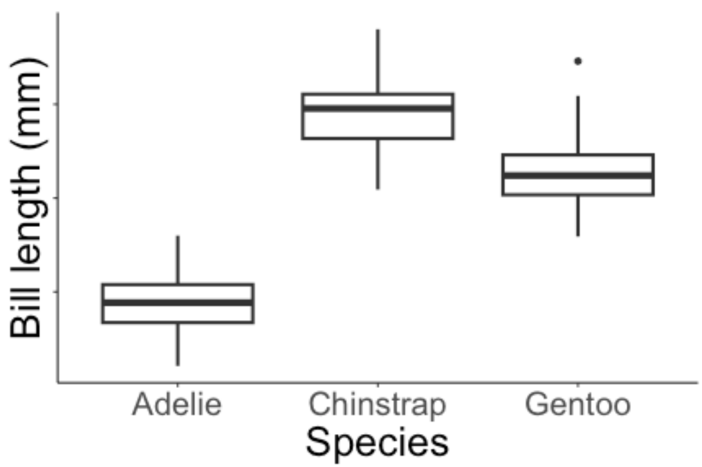
$$\text{bill length} = \beta_0 + \beta_1 \text{IsChinstrap} + \beta_2 \text{IsGentoo} + \varepsilon$$

> What do I need to test these hypotheses?

Need a test statistic!

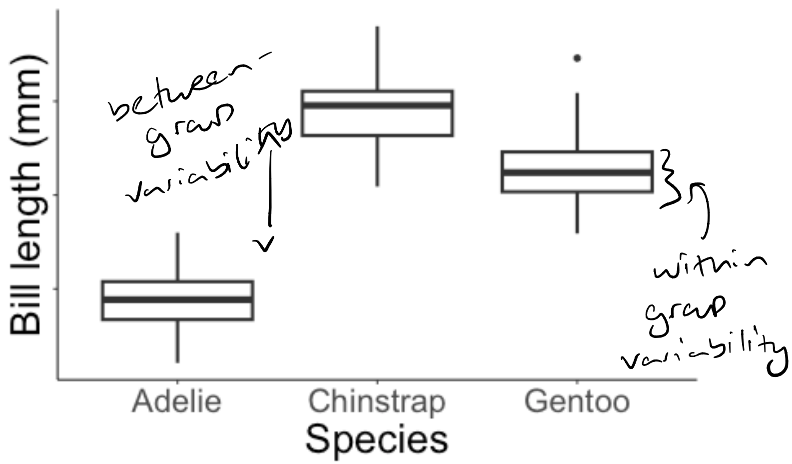
Intuition

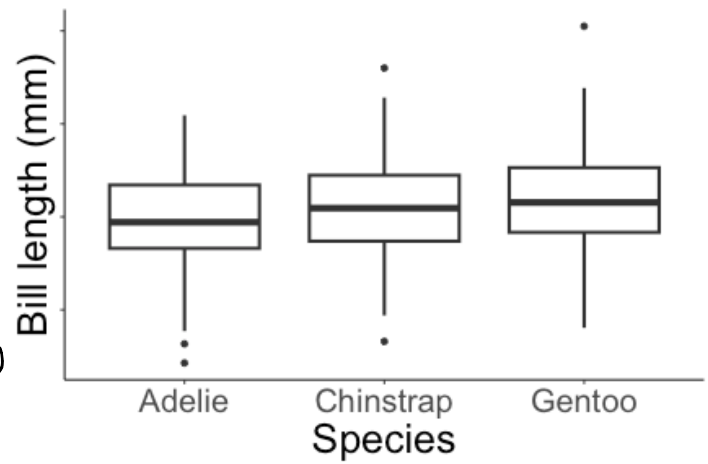
> Which of these two plots would show more evidence of a relationship between species and bill length?



Intuition

More evidence for a relationship when *between-group* variability is larger than *within-group* variability.



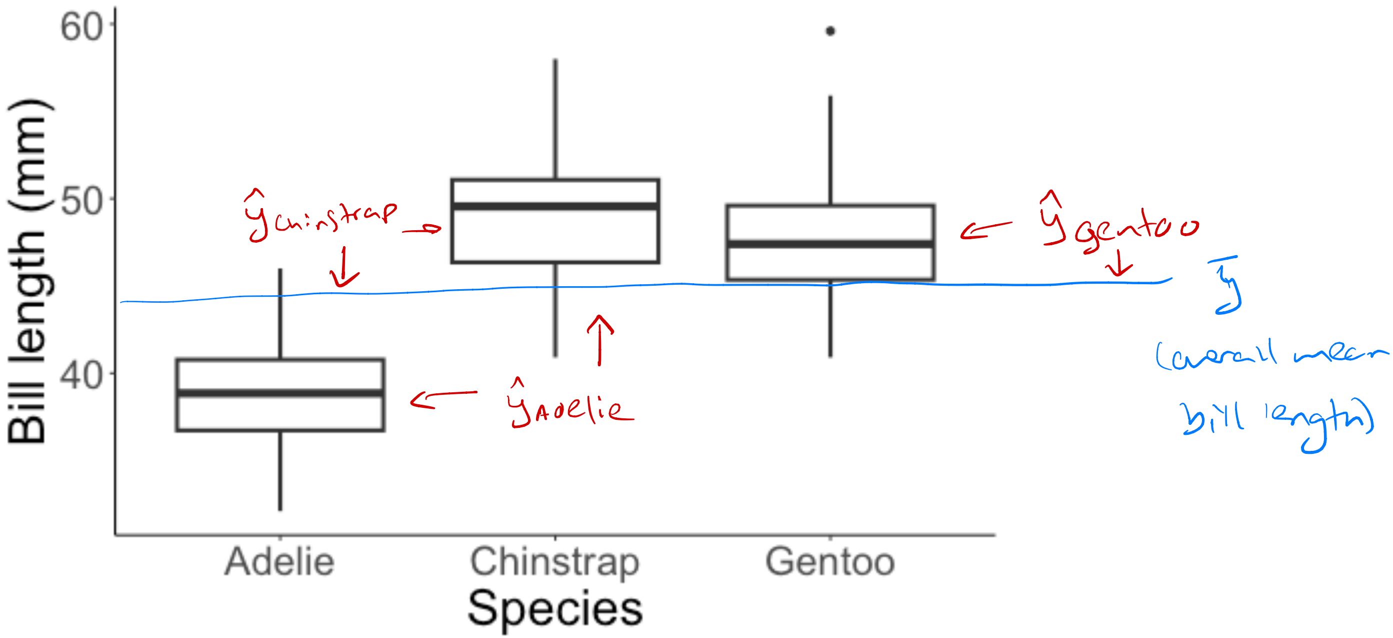


Large between-group variability, relative to within-group variability.

Small between-group variability, relative to within-group variability.

Test Statistic: $\dfrac{\text{between-group variability}}{\text{within-group variability}}$

Between-group vs. within-group variability



Between-group variability: $\sum_{i=1}^{n} (\hat{y}_i - \bar{y})^2$

Within-group variability: $\sum_{i=1}^{n} (y_i - \hat{y}_i)^2$ SSE!

Partitioning variability and degrees of freedom

total amount of in response

Variability:

$$\sum_{i=1}^{n}(y_i - \overline{y})^2 = \sum_{i=1}^{n}(\hat{y}_i - \overline{y})^2 + \sum_{i=1}^{n}(y_i - \hat{y}_i)^2$$

$$SSTotal \quad = \quad SSModel \quad + \quad SSE$$

(variability between groups)

(variability w/in groups)

Degrees of freedom:

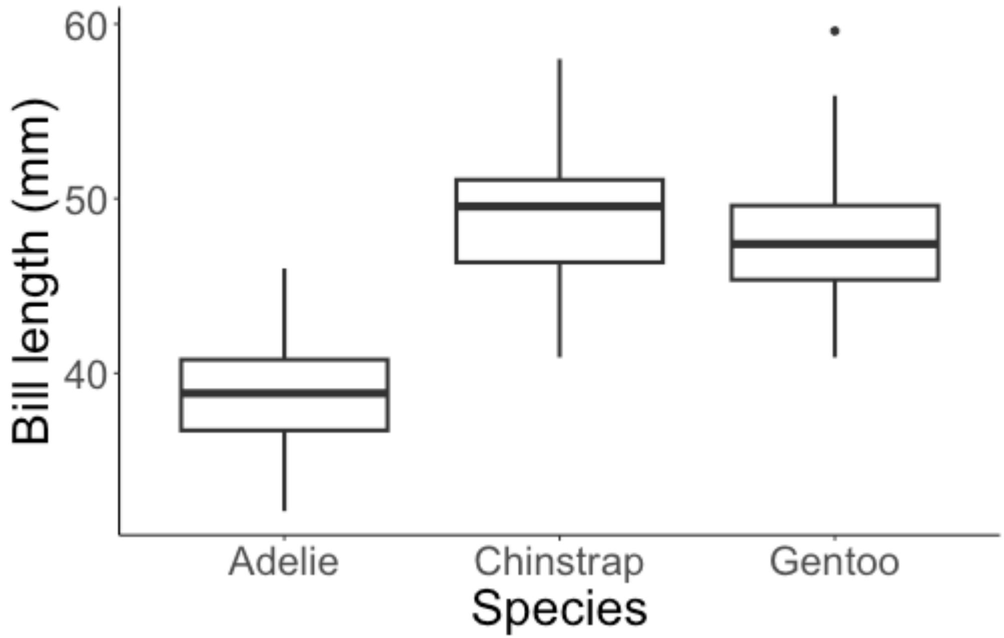
$$n-1 \quad = \quad (p-1) \quad + \quad (n-p)$$

n = # observations

p = # parameters in model (# of βs)

lose df by estimating mean of Y

Between-group vs. within-group variability



Test statistic:

$$F = \frac{\text{between group variance}}{\text{within group variance}} = \frac{SSModel/(p-1)}{SSE/(n-p)}$$

Partitioning variability

Variability:

$$\sum_{i=1}^{n}(y_i - \overline{y})^2 = \sum_{i=1}^{n}(\hat{y}_i - \overline{y})^2 + \sum_{i=1}^{n}(y_i - \hat{y}_i)^2$$

$$SSTotal \quad = \quad SSModel \quad + \quad SSE$$

Test statistic:

$$F = \frac{\frac{1}{p-1}\sum_{i=1}^{n}(\hat{y}_i - \overline{y})^2}{\frac{1}{n-p}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2} = \frac{\frac{1}{p-1}SSModel}{\frac{1}{n-p}SSE} = \frac{MSModel}{MSE}$$

MS = "mean squares"

comes from F distribution (under H_0)

Analysis of variance

Test statistic:

$$F = \frac{\frac{1}{p-1}\sum_{i=1}^{n}(\hat{y}_i - \overline{y})^2}{\frac{1}{n-p}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2} = \frac{\frac{1}{p-1} SSModel}{\frac{1}{n-p} SSE} = \frac{MSModel}{MSE}$$

Analysis of variance (ANOVA) table:

Source	df	SS	MS	F
Model	$p-1$	$SSModel$	$MSModel = \frac{SSModel}{p-1}$	$F = \frac{MSModel}{MSE}$
Residual	$n-p$	SSE	$MSE = \frac{SSE}{n-p}$	
Total	$n-1$	$SSTotal$		

Example

$n = 333$ penguins

$p = 3$ $(\beta_0, \beta_1, \beta_2)$

$$\sum_{i=1}^{n}(\widehat{\text{bill length}_i} - \overline{\text{bill length}})^2 = 7015.4$$ ← SSModel

$$\sum_{i=1}^{n}(\text{bill length}_i - \widehat{\text{bill length}_i})^2 = 2913.5$$ ← SSE

7015.4/2

Source	df	SS	MS	F
Model	2	7015.4	3507.7	397.25
Residual	330	2913.5	8.83	
Total	332	9928.9		

(n-1)

7015.4 + 2913.5

$\frac{2913.5}{330}$

$\frac{3507.7}{8.83}$

Class activity

Work on the class activity (handout).

Class activity

$n = 351$

$p = 8$

Work on the class activity (handout).

$7(15.49) = 108.43$

Source	df	SS	MS	F
Model	7	108.43	15.49	1.88
Residual	343	2826.32	8.24	
Total	350	2934.75		

$\frac{15.49}{8.24}$

$n-1$

SSmodel + SSE

$343(8.24)$